MODULE 31 · ALL DOMAINS

Confusion Pairs

The look-alike choices the exam sets side by side, the one detail that separates each pair, and practice at flipping the answer by changing that detail.

DOMAINS All 8 PAIRS 18 side by side MODULE 31 of 36

admin@sthithapragna.com

Identical twins look the same until you learn the one detail that tells them apart. **Most distractors are twins of the right answer.**

admin@sthithapragna.com

Architecture twins

PAIR	FIRST, WHEN	SECOND, WHEN
Workflow / agent	The steps are known in advance	The steps depend on what is found
Own loop / Agent SDK	A short loop you fully control	You want the agent loop built in
Agent SDK / Managed Agents	You run the process and sandbox	Anthropic hosts it; ZDR not needed
Client / server tool	Your code or data runs it	Anthropic runs it: web search

THE TIEBREAKER

The deciding details: who knows the steps, who runs the process, where the code lives.

Extension twins

PAIR	FIRST, WHEN	SECOND, WHEN
Custom tool / MCP server	One application needs it	Several apps or clients reuse it
Tool / Skill	An action or a data fetch	Know-how and procedure, loaded on demand
Permission rule / hook	A fixed allow, ask or deny	Logic that runs on an event
Prompt instruction / hook	Guidance that needs judgement	A rule that must always hold

THE TIEBREAKER

Reuse decides tool versus server. Must-always decides prompt versus code.

Delivery and cost twins

PAIR	FIRST, WHEN	SECOND, WHEN
Streaming / Batch	A person is waiting	Nobody waits; half price, within 24h
Prompt / response caching	Same prefix, new question	Identical request, reuse the answer
Effort / model tier	Same model, deeper reasoning	Still short at high effort
Adaptive / extended thinking	Current models decide the depth	Older 4.5 models: budget_tokens

THE TRAP

Response caching is yours to build. Prompt caching only discounts a repeated prefix.

Context and output twins

TECHNIQUE	USE WHEN	NOT WHEN
Structured outputs	Code must parse every reply	Free prose for a person
Prompt-only JSON	A quick prototype	Anything that must parse
Pruning (context editing)	Old tool results are spent	Every turn still matters
Compaction	A long conversation must continue	Only tool output is bloated
Isolation (subagents)	A side task needs its own window	Steps share heavy state

THE TIEBREAKER

Name what is filling the window, then pick the technique that removes exactly that.

Plumbing twins

PAIR	FIRST, WHEN	SECOND, WHEN
stdio / Streamable HTTP	Local process, one client	Remote, shared, OAuth
SSE / WebSocket	Claude API streaming	Your own two-way front end
Dated ID / dateless ID	Before 4.6: pin the date	4.6 on: the ID is the snapshot
Project / local / user	Shared, committed to the repo	Personal: one repo, or all

CURRENCY CHECK

The Claude API streams over SSE only. It has no WebSocket endpoint.

Separating twins under exam pressure

- **Find the deciding detail:** who waits, who runs it, how many reuse it, must it always hold.
- **Read the qualifier:** first, best, most cost-effective, least effort.
- **Both twins are real features.** Only one fits this situation.
- **Watch for the third twin:** a familiar fix from the wrong layer.
- **Practise the flip:** change one detail and name the new answer.

> ON THE EXAM
>
> The stem always holds the tiebreaker. If you cannot find it, re-read the last sentence.

> An insurer handles each claim the same way: extract the fields, check them against the policy, then draft a decision letter for a person to approve.
>
> A teammate proposes an autonomous agent with every tool available.

Which design fits best?

A An autonomous agent with every tool, as proposed

B A workflow that chains the three steps in code

C A team of agents, one supervising the others

D One long prompt that does all three steps at once

◆ PAUSE AND DECIDE

Which design fits best?

A An autonomous agent with every tool, as proposed

Autonomy the task does not need.

B A workflow that chains the three steps in code

Known steps: a workflow.

C A team of agents, one supervising the others

Even more machinery, same fixed path.

D One long prompt that does all three steps at once

Harder to test and validate per step.

THE FLIP

If each claim needed different steps depending on what was found, an agent would fit.

> A small start-up wants a long-running coding agent with a sandbox. It has no operations team to run containers.
>
> Its data has no zero-data-retention or residency requirement.

Where should the agent run?

A. A hand-written Messages API loop on its own servers

B. The Agent SDK in containers the start-up manages

C. Claude Managed Agents, hosted by Anthropic

D. Claude Code on one developer's laptop

◆ PAUSE AND DECIDE

Where should the agent run?

A A hand-written Messages API loop on its own servers

Builds the harness and hosting itself.

B The Agent SDK in containers the start-up manages

Needs the ops team it lacks.

C Claude Managed Agents, hosted by Anthropic

Hosted harness and sandbox.

D Claude Code on one developer's laptop

Not a production deployment.

THE FLIP

If zero data retention or a private network were required, self-host the Agent SDK.

> A research assistant must answer with current public information from the web.
>
> The team plans to build its own scraping tool and run it on its servers.

What should the team use?

A Anthropic's server-side web search tool

B A client tool wrapping a third-party scraping service

C An MCP server the team writes for web scraping

D Search results pasted into the system prompt nightly

◆ PAUSE AND DECIDE

What should the team use?

A Anthropic's server-side web search tool
The platform already does it.

B A client tool wrapping a third-party scraping service
Builds what the platform provides.

C An MCP server the team writes for web scraping
Even more to build and run.

D Search results pasted into the system prompt nightly
Stale by morning, and bloats context.

THE FLIP

If the data sat behind the company firewall, a client tool or MCP server would fit.

> Analysts want Claude to follow the firm's twelve-step quarterly reporting procedure, with its templates and checks.
>
> The procedure is long and is needed a few times a quarter.

How should the team provide it?

A Paste the procedure into every system prompt

B Package the procedure and templates as a Skill

C Build a custom tool that returns the procedure text

D Fine-tune a model on past quarterly reports

◆ PAUSE AND DECIDE

How should the team provide it?

A Paste the procedure into every system prompt
Pays for it on every request.

B Package the procedure and templates as a Skill
Know-how, loaded on demand.

C Build a custom tool that returns the procedure text
A tool is for actions and data.

D Fine-tune a model on past quarterly reports
Training, where packaging solves it.

THE FLIP

If Claude needed live sales figures for the report, that part is a tool.

> Claude Code must never run terraform apply in any company repository.
>
> A line in CLAUDE.md saying so was ignored once last month.

What is the best control?

A. Reword the CLAUDE.md line in capitals, with a reason

B. Set every Bash command to ask for approval

C. A deny rule for the command in managed settings

D. A PostToolUse hook that logs every apply it sees

◆ PAUSE AND DECIDE

What is the best control?

A Reword the CLAUDE.md line in capitals, with a reason

Still guidance, not enforcement.

B Set every Bash command to ask for approval

Approval fatigue for every command.

C A deny rule for the command in managed settings

A fixed rule, enforced everywhere.

D A PostToolUse hook that logs every apply it sees

Logs it after it has already run.

THE FLIP

If apply were blocked only for production workspaces, a PreToolUse hook could decide.

> A software company imports about 80,000 new sign-ups every Monday and wants a personalised onboarding email for each.
>
> The emails must be ready by Tuesday morning.

How should it generate them?

A Stream each email as it is generated

B Real-time calls with many more parallel workers

C Real-time calls with fast mode turned on

D The Message Batches API, submitted on Monday

◆ PAUSE AND DECIDE

How should it generate them?

A Stream each email as it is generated
Nobody is watching the stream.

B Real-time calls with many more parallel workers
Full price, plus rate-limit pressure.

C Real-time calls with fast mode turned on
Pays more for speed nobody needs.

D The Message Batches API, submitted on Monday
Nobody waits: half price.

THE FLIP

If each new user waited on screen for their own email, stream it.

> A help-centre bot receives thousands of identical questions a day, such as "What are your opening hours?", with identical answers.
>
> It already uses prompt caching on its long system prompt.

What would cut cost most for these questions?

A Nothing more, since prompt caching already covers it

B Your own cache of answers to identical requests

C The Message Batches API for every question

D A move to the cheapest model for all traffic

◆ PAUSE AND DECIDE

What would cut cost most for these questions?

A Nothing more, since prompt caching already covers it

Still pays for every reply's output.

B Your own cache of answers to identical requests

No call at all for repeats.

C The Message Batches API for every question

Users are waiting for answers.

D A move to the cheapest model for all traffic

Risks quality on harder questions.

THE FLIP

If every question differed but shared the long prompt, prompt caching is the answer.

> A triage feature runs Sonnet 5.5 at low effort to save tokens. Accuracy on hard cases is below the target.
>
> A teammate proposes moving to Opus 5.5.

What should the team try first?

A Move to Opus at its default effort, as proposed

B Move to Fable, the most capable model

C Re-run the evals on Sonnet at higher effort

D Add retries until hard cases come out right

◆ PAUSE AND DECIDE

What should the team try first?

A Move to Opus at its default effort, as proposed

A bigger change than the evidence asks.

B Move to Fable, the most capable model

Even bigger and costlier.

C Re-run the evals on Sonnet at higher effort

Same model, deeper reasoning first.

D Add retries until hard cases come out right

Retries do not add judgement.

THE FLIP

If Sonnet still fell short at high effort on the evals, move up a tier.

> A coding agent's context fills with old file-read results it no longer needs.
>
> The conversation itself is short, and every turn still matters.

What should the developer use?

A Context editing to clear old tool results

B Compaction of the whole conversation

C A fresh subagent for every file the agent reads

D A model with a larger context window

◆ PAUSE AND DECIDE

What should the developer use?

A Context editing to clear old tool results
Removes exactly what is spent.

B Compaction of the whole conversation
Summarises turns that still matter.

C A fresh subagent for every file the agent reads
Far more machinery than needed.

D A model with a larger context window
More room for the same clutter.

THE FLIP

If a long conversation itself had to continue, compaction would fit.

> A team is building an MCP server for its ticketing system. It will run centrally and serve 40 engineers' Claude Code sessions and two web apps.
>
> Each user must see only their own team's tickets.

Which two choices fit? Choose two.

A The stdio transport, launched by each client

B Pass each user's ticketing token straight through

C The Streamable HTTP transport, deployed once

D WebSocket as the standard MCP transport

E OAuth for authorisation on the HTTP transport

◆ PAUSE AND DECIDE

Which two choices fit? Choose two.

A The stdio transport, launched by each client
Local, one client at a time.

B Pass each user's ticketing token straight through
Token passthrough is forbidden.

C The Streamable HTTP transport, deployed once
Remote and shared.

D WebSocket as the standard MCP transport
Not a standard transport.

E OAuth for authorisation on the HTTP transport
Per-user identity, done right.

THE FLIP

For one developer's local server reading local files, stdio is the answer.

Module 31 in five lines

- **Architecture:** known steps, workflow; who hosts decides SDK or Managed Agents.
- **Extensions:** reuse means MCP; know-how means Skill; must-always means code.
- **Delivery and cost:** waiting means stream; identical means your own cache.
- **Context:** prune spent results, compact long chats, isolate side tasks.
- **Plumbing:** stdio local, Streamable HTTP shared; SSE, not WebSocket, for Claude.

COMING UP

Next: everything that changed since the July 2026 guide, and the exam-safe answer for each.